

Demographic genetics of American beech (*Fagus grandifolia* Ehrh.) IV. Development of genetic variability and gene flow during succession in a coastal plain forest in Maryland

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Abstract

Genetic recovery of an American beech (*Fagus grandifolia*) population in deciduous forests that were once pastures was studied using 16 allozyme loci from 410 individuals in a 600 m × 600 m study plot in Maryland, USA. We also examined the spatio-temporal genetic structure of the American beech population at a regional scale. Overall genetic diversity of mature trees was measured by estimating average heterozygosity ($H = 0.156$). Rare alleles were observed in five loci, *Lap*, *6Pgdh3*, *Pgi*, *Adh1* and *Got3*. Mature individuals were divided into three size classes based on d.b.h. The genetic component of each size class was compared and it was revealed that several alleles (*Pgm-a*, *6Pgdh3-a* and *Lap-b*) were shared only in specific size classes. The spatial distribution of the genotypes demonstrated a conspicuous localization in three loci (*Aco*, *Adh1* and *Idh*). Spatial autocorrelation analyses were carried out among the mature trees for a 20 m interval, and were positive for 0–120 m and negative for >180 m. Distrograms indicated that a unique genetic localization occurs among mature individuals. Seven hundred and seventy-five seedlings in the 10 m × 120 m transect were analyzed to measure gene flow via seed and/or pollen. We obtained a genetic neighborhood area of 1.17 ha and an effective population size of 32.4. The temporal and spatial modes of genetic recovery of the population are discussed in the context of conservation biology.

Keywords: conservation biology, *Fagus grandifolia*, gene flow, isozyme, spatial autocorrelation.

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Introduction

It is important to understand the development of genetic variation in plant populations from a limited number of

founder individuals. The development of genetic diversity in populations is an important issue in conservation biology. The recovery of genetic diversity in a population that developed from a small number of founder individuals is influenced by several factors: the characteristics of the founder individuals, effective mating among individuals of a small population through effective pollen and gene flow, the presence of pollinating agents, seed dispersal, establishment of seedlings at suitable sites, and the potential for vegetative reproduction. The genetic

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consequences of fragmentation and isolation of populations have been studied for many plant species (Morgan 1999; Young *et al.* 1999, 2001; Knapp *et al.* 2001; Neel & Ellstrand 2001), but few studies have quantified changes in the patterns of genetic diversity following development of a population from a once fragmented habitat.

In a series of studies, we have examined the demographic genetic substructuring of American beech (*Fagus grandifolia* Ehrh.) (Kitamura *et al.* 1998, 2000, 2001, 2003; Kitamura & Kawano 2001), Siebold's beech (*Fagus crenata*) (Kawano & Kitamura 1997; Kitamura *et al.* 1997a,b; Ohkawa *et al.* 1998; Takenaka *et al.* 2002) and Korean beech (*Fagus multinervis*) populations (Ohkawa *et al.* 2006). These studies have demonstrated clear patterns of genetic differentiation in relation to demographic genetic substructuring within the populations examined (cf. Dietz *et al.* 1999; Little & Dale 1999; Li *et al.* 2001).

Based on our previous studies, we distinguished two types of American beech populations based on reproductive patterns (Kitamura & Kawano 2001). American beech populations of the eastern mountain ranges (e.g. Great Smoky and Blue Ridge Mountains), from the south to as far north as Quebec, Canada, develop root suckers and are capable of reproducing by both sexual and asexual methods (Kitamura *et al.* 2000, 2001). In contrast, the American beech populations of the south-eastern Coastal Plain as well as the Mississippi River Basin and Ozark Mountains are only capable of sexual reproduction (Kitamura *et al.* 1998, 2001, 2003; Kitamura & Kawano 2001).

American beech growing at the Smithsonian Environmental Research Center (SERC) in Maryland belong to the non-root suckering populations, and regenerate only from seed by obligate outcrossing (Kitamura *et al.* 1998). In a previous paper, we examined fine-scale genetic substructuring of a small portion of the American beech population in a small area at SERC (Kitamura *et al.* 2003). In this paper, we expanded our previous study by determining the genetic structure of mature individuals over a much broader area (600 m × 600 m). We also attempted to measure the degree of gene flow within the plot by applying spatial autocorrelation analyses to elucidate the differences, if any, in the degree of gene dispersal at both the seedling and the mature tree stages. Finally, we have discussed the implications of the present study in the context of conservation biology.

Materials and methods

Species description

The American beech, *Fagus grandifolia*, is monoecious (Brown 1921) and mast flowering and fruiting occurs at an interval of 2–3 years (Deen 1936). Its breeding system is obligatory outcrossing because the outcrossing rate is esti-

mated to be 1.0 (Kitamura *et al.* 1998). Seed dispersal is a typical barochory, and secondary dispersals are made by blue jays, *Cyanocitta cristata* (Johnson & Adkisson 1985), and small mammals (Rohrig & Ulrich 1991). American beech is tolerant of shade (Batista *et al.* 1998) and juveniles can survive more than 70 years of shade suppression (Poulson & Platt 1989; Canham 1990). American beech has two modes of regeneration, sexual reproduction by seed and asexual reproduction by root suckers (Ward 1961).

Study site

The present study was conducted at the property of the SERC (38°51'N, 76°32'W), on the Inner Coastal Plain of Maryland, USA. The forest canopy is composed of more than 30 species and the dominant species are *Fagus grandifolia*, *Liriodendron tulipifera*, *Quercus alba*, *Quercus falcata*, *Carya glabra* and *Carya tomentosa*. The forest understory has a well-developed herbaceous layer that includes *Claytonia virginiana*, *Dentaria laciniata*, *Arisaema triphyllum*, *Podophyllum peltatum*, *Smilacina racemosa*, *Geranium maculatum*, *Sanguinaria canadensis* and *Desmodium nudiflorum* (Whigham *et al.* 1993).

The study area was once a pasture, abandoned approximately 150 years ago, and has subsequently been undergoing secondary succession (Tibbs 1995). Some of the largest individuals of American beech were clearly present at the time of abandonment, and this founder population is likely to be the source of the propagules that have colonized the site since then. There is, however, no method to determine whether any juvenile populations were present within the area 150 years ago.

The study plot (Fig. 1) at present is dominated by American beech (*F. grandifolia*). The study plot includes one area (shown as the blank space in the upper right portion of Fig. 1) that was a farm field until it was abandoned to secondary succession in the 1970s. In 1998, all trees with a d.b.h. ≥ 20 cm were mapped and their diameters were measured (Fig. 1) (G. Parker, pers. comm., 1998).

The leaves from 410 mature individuals were sampled in the study plot for enzyme extraction. We collected leaves from all 351 beech trees larger than 40 cm d.b.h. We also sampled leaves from all 59 individuals with a d.b.h. ≥ 20 cm that occurred in a 300 m × 300 m plot (located in the lower left quadrant of Fig. 1) within the 600 m × 600 m plot.

Data for all individuals within the 10 m × 120 m plot had previously been sampled and analyzed (Kitamura *et al.* 2003). This transect was established in May–June 1994, the season following masting in 1993. We used the seedling data from this transect to determine seed and pollen dispersal.

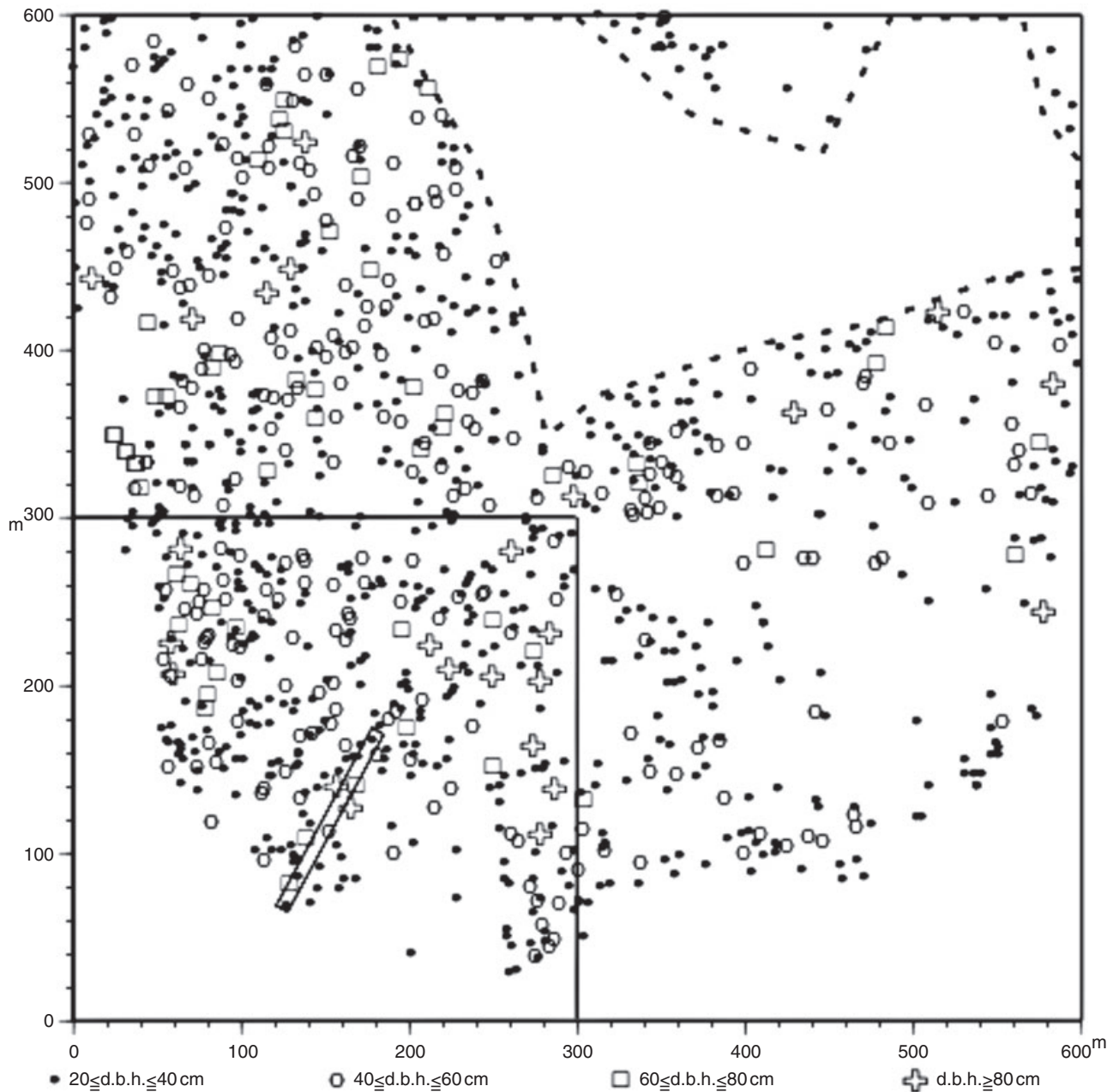


Fig. 1 The 600 m × 600 m study plot dominated by American beech trees (with d.b.h. ≥ 20 cm) at the Smithsonian Environmental Research Center (SERC), Maryland, USA. One 10 m × 120 m study plot (Kitamura *et al.* 2003) and its surrounding 300 m × 300 m area is shown in the lower left corner. The open zone surrounded by the dotted line in the upper right corner indicates an area that was once a corn field (Tibbs 1995) and is now covered by an oak forest.

Allozyme analysis

Extraction, electrophoreses and staining of enzyme activities were carried out as described previously (Kitamura & Kawano 2001; Kitamura *et al.* 2003).

Sixteen loci from 12 enzyme systems were analyzed: 6Pgdh1, 6Pgdh2, 6Pgdh3, Aco, Acp, Adh1, Adh2, Dia, Fum, Got1, Got3, Idh, Lap, Mdh, Pgi and Pgm. Except for one

monomorphic locus (*Adh2*), the remaining 15 loci were polymorphic.

Differentiation among size classes of mature trees

From the size of the trees, it appeared that the American beech population in the plot included several generations.

To examine whether or not there is any genetic differentiation among mature individuals, the trees were divided into three d.b.h. size classes: (i) 273 individuals, $40 \leq \text{d.b.h.} < 60 \text{ cm}$; (ii) 55 individuals, $60 \leq \text{d.b.h.} < 80 \text{ cm}$; and (iii) 23 individuals, $\text{d.b.h.} \geq 80 \text{ cm}$. The frequencies of alleles, genotypes, homozygote and heterozygote, as well as heterozygosity and Wright's F -statistics for each locus were calculated and compared among the size classes. Heterozygosity was calculated according to the equation by Nei and Roychoudhury (1974),

$$\hat{h} = \frac{2n}{2n-1} \left(1 - \sum \hat{x}_i^2 \right)$$

Wright's F -statistics (Wright 1943, 1951, 1965) were calculated according to Weir and Cockerham (1984) using the computer program FSTAT (Goudet 1995) for the different size classes.

Because of the difference in the sampling area for the $20 \text{ cm} \leq \text{d.b.h.} < 40 \text{ cm}$ size class, we excluded the 59 individuals collected only within $300 \text{ m} \times 300 \text{ m}$ when comparing the genetic diversity among the size classes.

Spatial autocorrelation among mature trees

If the current American beech population in the SERC plot developed from a limited number of founder trees, the spatial genetic structures of the individuals within the plots must have been influenced by them during the course of secondary succession. Spatial autocorrelation (Sokal & Oden 1978a,b) was calculated for each allele to examine the large-scale spatial genetic substructuring:

$$I = n \sum_{ij} w_{ij} z_i z_j / W \sum_{i=1}^n z_i^2$$

Some loci were omitted from the spatial autocorrelation analysis because only a small number of individuals (1–3) showed polymorphisms: *6Pgdh1*, *Got1*, *Got3*, *Mdh* and *Dia*. Significance levels were calculated using the jackknifing method. Computations were carried out at the Computer Center for Agriculture, Forestry and Fisheries Research, Ministry of Agriculture, Forestry and Fisheries, Japan.

Estimation of genetic neighborhood area and effective population size

Data used in the estimation of gene flow were obtained from a study of 873 seedlings sampled in a $10 \text{ m} \times 120 \text{ m}$ plot (Kitamura *et al.* 2003), and from the 410 mature trees sampled in the present study. As four loci (*Aco*, *Dia*, *Idh*

and *Mdh*) were not analyzed in the seedling study, combinations of multilocus genotypes (MLG) for 11 polymorphic loci of mature trees were used to determine the prime parents of each seedling (Kawano & Kitamura 1997).

To search for the parent candidates of each seedling, all combinations of multilocus genotypes, which could be produced by currently present fertile trees, were compared with currently existing genotypes of seedlings. The nearest mature individuals among the possible combinations that can produce seedlings with the designated multilocus genotypes are assumed to be possible parental candidates. It is reasonable to assume that the seed dispersal ranges are smaller than the pollen dispersal ranges in American beech. As for the distances between the seedlings and the two parental candidates, we recognized the nearer mature trees as seed donors and the others as pollen donors.

When the prime parents of all the seedlings were determined, the variance in the dispersal distance (σ^2) (Gliddon *et al.* 1987), the genetic neighborhood area (A) (Wright 1943, 1951; Silvertown & Lovett Doust 1993) and the effective population size (N_e) (Crawford 1984; Silvertown & Lovett Doust 1993) were calculated using the outcrossing rate $t = 1$ (Kitamura *et al.* 1998). The following equations were used to calculate σ^2 , A and N_e :

$$\sigma^2 = \frac{t}{2} \sigma_{\text{pollen}}^2 + \sigma_{\text{seed}}^2$$

$$A = 4\pi\sigma^2$$

$$N_e = A \frac{N}{2} (1 + t)$$

Theoretically, when the population is stable and stationary, and the distribution per parent is Poisson (Kimura & Crow 1964), the probability of the existence of the parents within the radius of ' $r\sigma$ ' is $P = 1 - e^{-\frac{r^2}{2}}$. It has been stated that N_e is equivalent to the number of reproducing individuals in a circle with a radius of 2σ (Wright 1943, 1946, 1951). Therefore, the probability of the existence of parents within a radius of 2σ ($r = 2$) is $P = 1 - e^{-2} = 0.865$ (Crawford 1984). Therefore, we used 86.5% of the seedlings to estimate the gene dispersal variance, σ^2 , the genetic neighborhood size, A , and the effective population size, N_e .

Technically, we estimated variances for the Euclidian distances of the parent candidates using 755 seedlings, which is 86.5% of 873 seedlings. We assumed that the density of reproductive plants in the population, N , is equal to the density of mature trees obtained in the present study.

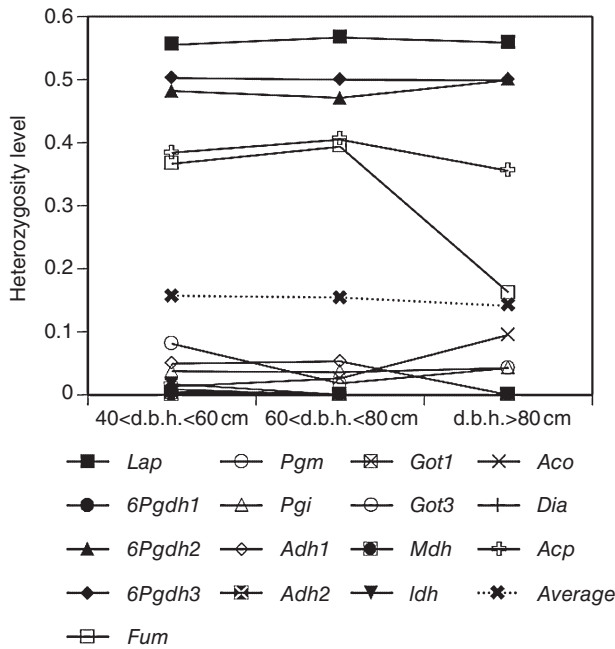


Fig. 2 Levels of heterozygosity for each locus in the three different size classes of mature American beech.

Results

Isozyme polymorphism among the size classes

Allele and genotypic frequencies, Nei's estimation of heterozygosity and Wright's *F*-statistics are given in Appendices I, II and III, respectively.

These data describe the genetic structures of the American beech population in the three different size classes. Heterozygosity is high in *Lap*, *6Pgdh2*, *6Pgdh3*, *Acp* and *Aco* (Fig. 2). For the two smallest size classes, polymorphism was low for all other allozymes. The average heterozygosity did not differ among the three size classes.

Fan diagrams (Fig. 3) show the genetic polymorphisms for each size class for the representative loci. Allelic components were not different among size classes in *Acp*, *Aco*, *Adh1*, *Fum* (Fig. 3a) and *6Pgdh2*; however, genotype polymorphism was high in *Fum* (Fig. 3a) in the smallest size class. The *d*-allele in *Pgi* was found only in the smallest and the largest size classes (Fig. 3b). In *Pgm*, *6Pgdh3* (Fig. 3c) and *Lap*, rare alleles and genotypes were observed only in the smallest size class.

Spatial genotypic substructures within the study plot

Four examples of the spatial distribution of genotypes of mature trees are demonstrated in Figure 4. A high level of genotypic polymorphism was observed in *Acp*, *Fum*, *Lap*, *6Pgdh3* (Fig. 4a) and *6Pgdh2* throughout the plot. Some genotypes of *Aco*, *Adh1* (Fig. 4b) and *Idh* were localized

within the plot. Genetic diversity was low in *Dia*, *Got1* (Fig. 4c), *Got3*, *Mdh* and *6Pgdh1*, whereas spatial substructuring was absent in *Pgi* (Fig. 4d) and *Pgm*.

Spatial autocorrelation analyses

We used classes with a minimum 20 m interval distance to calculate the spatial autocorrelation (Moran's *I*) among mature individuals in the plot. Distrograms of *I*(20) for 10 loci are shown in Figure 5, and the significance values are provided in Appendix IV. Most of the significant positive values of Moran's *I* were observed in distance classes of less than 120 m, whereas most of the significant negative values were observed in distance classes greater than 180 m (Appendix IV).

Genetic subdivisions of the population

The exclusion probability was calculated as $Q = 0.874$ when determining the parental candidates (Grundel & Reets 1981). The Euclidian distance used in the calculation of effective seed dispersal and pollen flow distances is demonstrated in Figure 6.

We estimated the seed dispersal distance to be $2\sigma_{\text{seed}} = 41.85$ m and the pollen flow distance to be $2\sigma_{\text{pollen}} = 62.64$ m. The calculated gene flow variance, σ^2 , was 928.33 and the genetic neighborhood size, *A*, was 1.17 ha.

To estimate the effective population size, we first determined the area of the plot containing American beech. For this calculation we did not include the younger forest (the blank area in the upper part of the plot in Fig. 1) or the areas around the margin of the plot that were not forested or did not include any American beech. The density of American beech individuals with a d.b.h. >30 cm and that were capable of producing seeds as well as pollen was estimated to be 27.8 individuals/ha. From this, we estimated the effective population size, *N_e*, of the American beech in the plot to be 32.4 mature individuals.

Discussion

Local populations of perennial plants include individuals of various ages, resulting in a unique age structure. In populations of long-lived trees, individuals of the same size can be quite different in age because of differences in growing conditions over long periods of time. For example, many tree populations include so-called 'oskars', which are very old individuals that have stunted growth because they have been growing under suppressed conditions (Silvertown 1983). Thus, because of differences in age and growth, populations of tree species have different size-structure and age-structure characteristics. Most studies of tree populations focus on size-

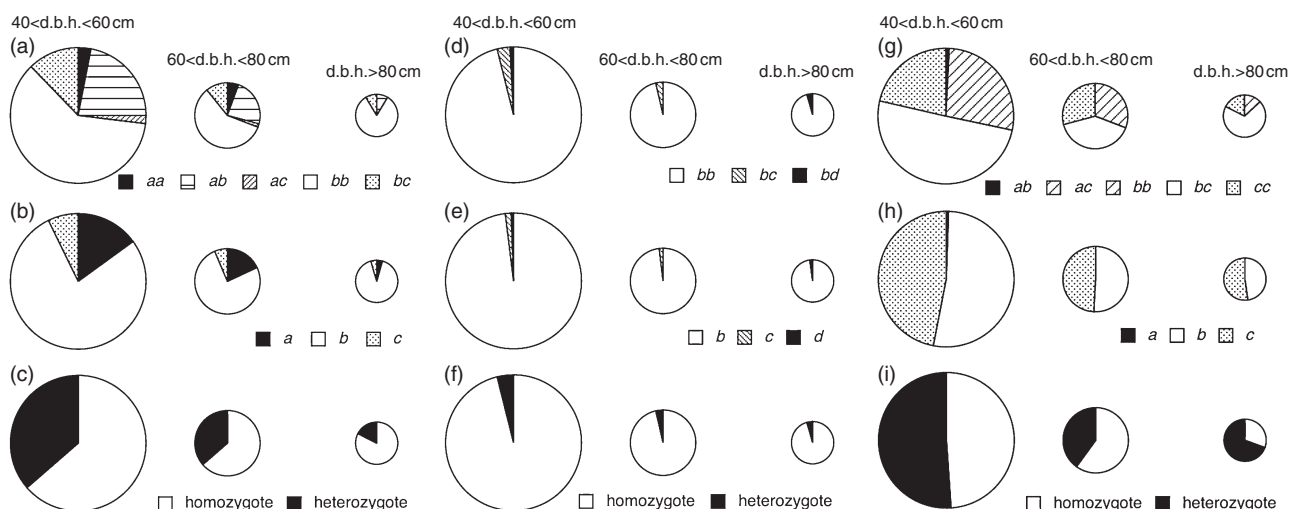


Fig. 3 Fan diagrams illustrating the frequencies of different genotypes, alleles, homozygotes and heterozygotes for the three different size classes. The differences in the size of the fan diagrams are relative to the numbers of individuals included. (a–c) *Fum*, (d–f) *Pgi* and (g–i) *6Pgdh3*. (a,d,g) genotype frequencies, (b,e,h) allele frequencies and (c,f,i) homozygote and heterozygote frequencies.

structure because there is no method for determining the exact ages of individuals without destructive sampling or collecting cores to determine the age from an analysis of annual growth rings. In addition, the age-structure and size-structure of a local population is influenced by patterns of recruitment, particularly in tree gaps, which leads to a mosaic of individuals of varying sizes and ages. Recruitment over time also results in a mosaic of genetic structures in both space and time.

The SERC forest has characteristics of a complex population, a result of historical events. There are no historical documents that describe the history of the SERC forest, but we were able to reconstruct part of the history based on the sizes and shapes of the largest trees in the forest (D. Whigham and G. Parker, pers. obs., 1995) and information compiled from an analysis of tree cores (Tibbs 1995). Most forests in the area have been logged and many were converted to agricultural fields. Other forests were only partially cleared and were used for grazing cattle. Based on the presence of very large trees (mostly American beech or oaks) growing in the open, we believe that most of the study area was a pasture in colonial times. One small portion, however, was used as an agricultural field until the mid 1950s. The younger successional forest in that area had large beech trees. In contrast, the rest of the study area has beech trees of all sizes and age classes, indicating that there has been a gradual recovery of genetic variation from the founder populations that would have been present at the time the pasture was abandoned.

Given that it takes approximately 50 years for a newly established American beech to become large enough (i.e. approximately 30 cm in d.b.h.) to flower and produce seeds, individuals established from the founder trees that

were present when the site was a pasture would have most likely produced seeds over the past 100 years. Although we do not know the details of seedling establishment in this forest, our earlier work (Kitamura *et al.* 2003) clearly shows that American beech seedlings are quite abundant in the years following masting, and a good number of juvenile successors scattered over the woodland floor were observed. Thus, we can assume that seedling establishment occurred effectively in the gap sites on the forest floor during the period of development of the current forest population (Kitamura *et al.* 1998).

The level of genetic diversity found in the SERC population is very high and is comparable to the levels of diversity recorded in other natural stands of American beech (Kitamura *et al.* 2000, 2001, 2003; Kitamura & Kawano 2001). This result indicates that the founder population at the SERC site probably also had a relatively high level of genetic diversity and was a remnant of the original forest population that had high genetic diversity. It is noteworthy that rare alleles and genotypes exist within the mature individuals, for example, *Acp*, *Fum*, *Lap*, *6Pgdh3* and *6Pgdh2*; this also indicates that successive generations are likely to maintain high genetic diversity. There is some evidence of substructuring within the American beech population; however, the value for σ , that is, 'the panmictic circle of 2σ radius at the neighborhood's center', is 121.8 m (Wright 1946; Levin & Kerster 1971, 1974; Schaal & Levin 1978; Crawford 1984). This value is similar to the estimate from the 120 m distance based on the spatial autocorrelation analysis (Kitamura *et al.* 2003). These two estimates suggest that the unit of an efficient breeding population is approximately 120 m in the SERC forest. However, one question regarding the genetic

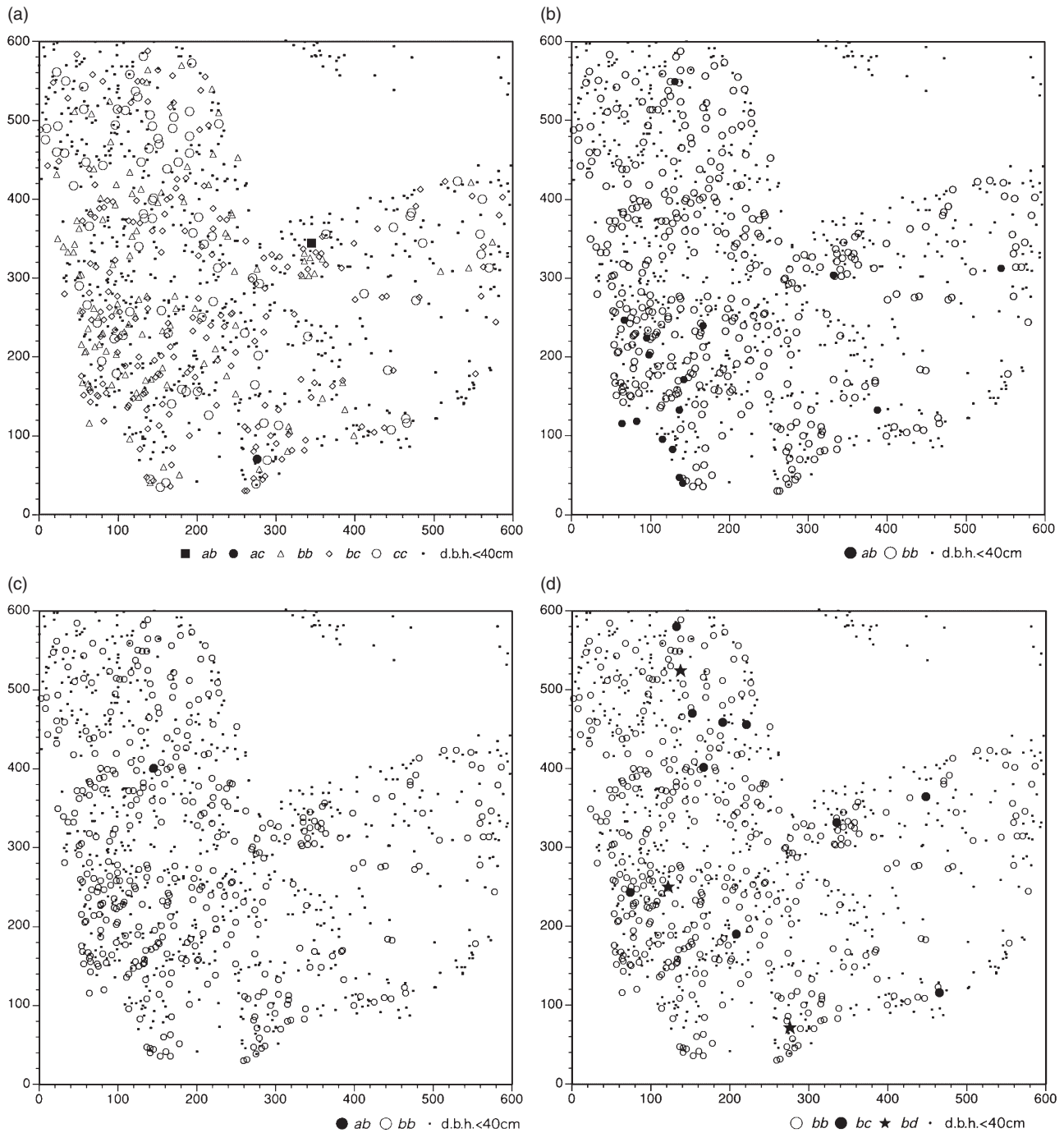


Fig. 4 Distributions of mature trees with d. b.h. ≥ 40 cm, with different genotypes for four loci. (a) *6Pgdh3*, (b) *Adh1*, (c) *Got1* and (d) *Pgi*.

variability of the SERC American beech population remains unanswered. Would a 150-year period be sufficient to reach a genetic equilibrium or would there indeed be an equilibrium state within a population?

Another feature of the SERC population is that it contains rare and/or remnant genetic variation. In long-lived woody species there is evidence regarding the existence

of infrastructural networks that offer the chance for these rare or remnant genotypes to persist or to become established as a result of effective gene flow within the local population networks.

Perhaps we should not ignore rare and/or remnant genetic variations currently included in the beech population, although they were excluded from the calculation

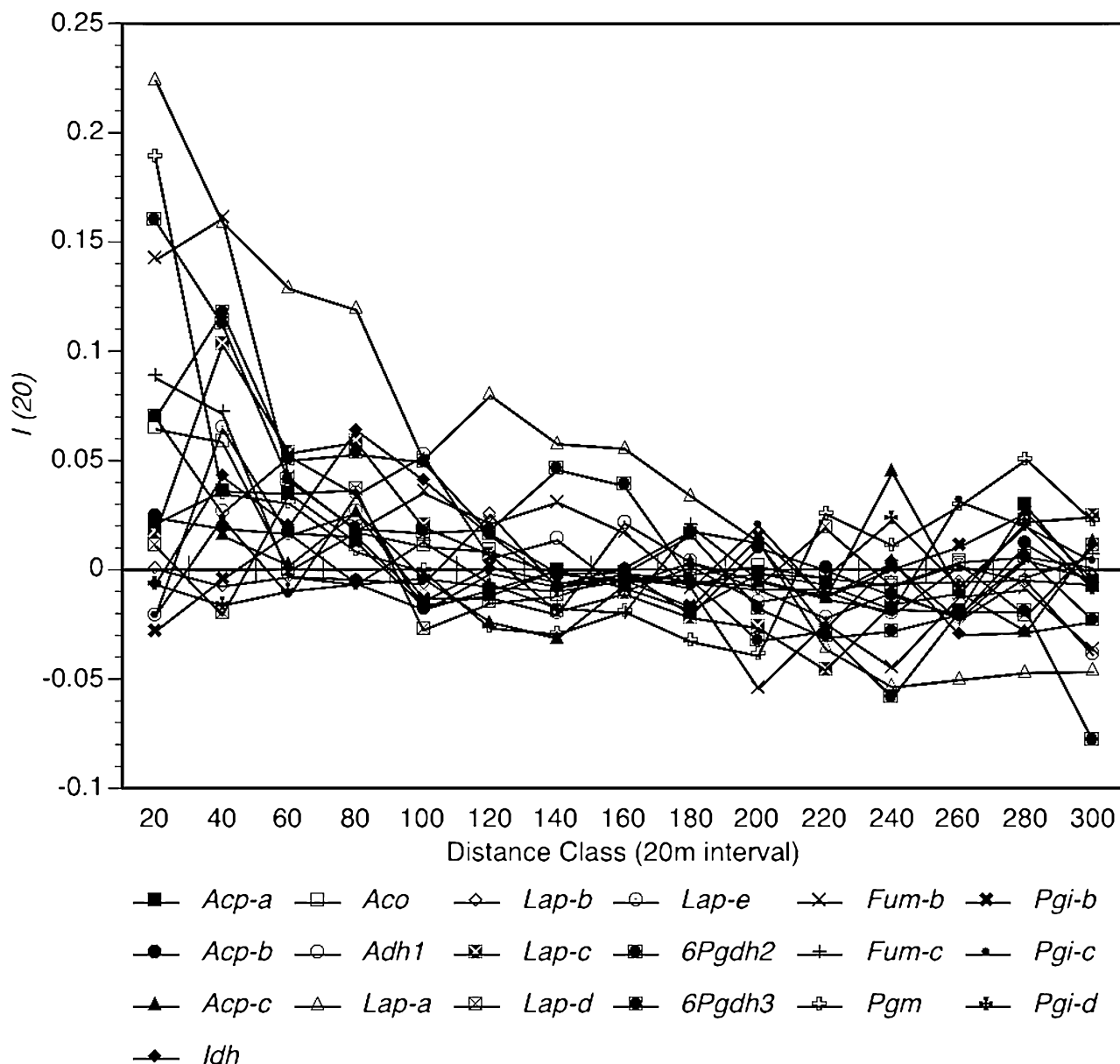


Fig. 5 Distograms showing spatial autocorrelation (Moran's I) for 10 loci. Distance classes were discriminated by 20 m intervals. Significance values are presented in Appendix IV.

of gene flow variance and effective population size by stochasticity because of their exceedingly low frequencies within the population. In the case of long-lived woody species, infrastructural networks might exist in localized individual assemblages of different ages, where these rare or remnant genotypes might have a chance to persist within a local population. Indeed, rare or remnant alleles appear in certain aged size classes. Although polymorphisms are present in very low frequencies within a population, they are important for the estimation of rare, but actual, long-distance gene flow within local population networks.

When discussing the local and regional scales of population structures, occasional gene flow between different cohorts must not be ignored. Effective population size can be considered to be an assemblage of individuals in which gene exchange is normally guaranteed and maintains a certain level of genetic polymorphism.

The genetic diversity of the SERC population suggests that the founder population was of adequate size for maintaining a certain level of genetic polymorphism. If the number of individual plants capable of exchanging genes is limited, the total genetic diversity within a population declines so rapidly that the genetic variation

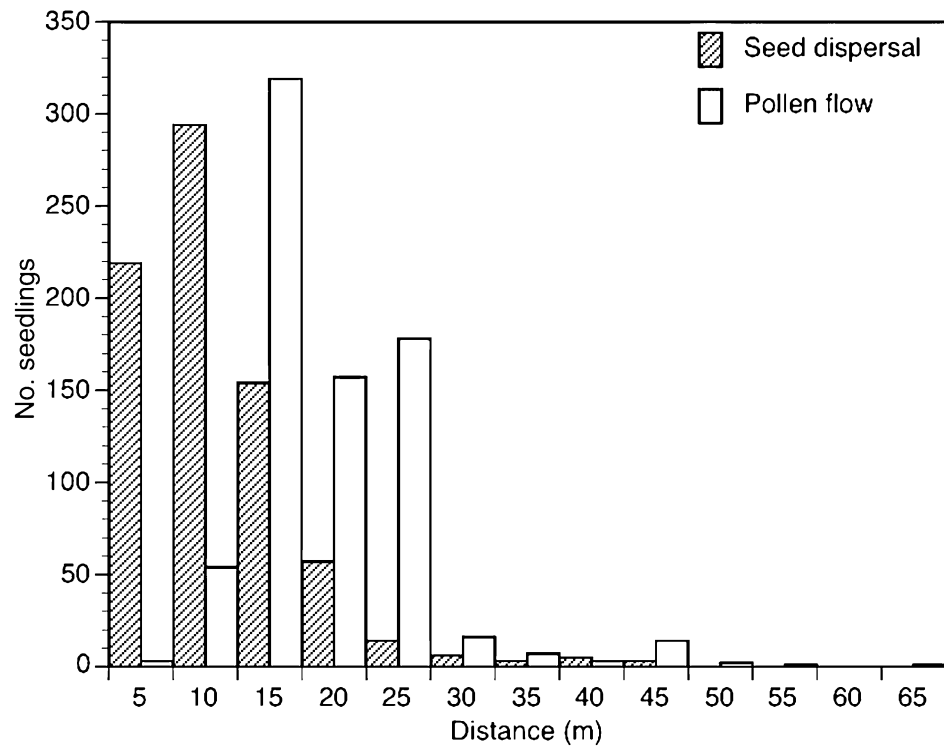


Fig. 6 Ranges of seed dispersal and pollen flow distances estimated by multilocus genotypes of 11 loci for 86.5% (755 individuals) of the seedlings sampled from a 10 m $\times$ 120 m transect.

normally maintained in that population can no longer be maintained. Examples of this have been found in small, isolated Siebold's beech populations scattered in the lowlands of Toyama, in central Honshu (Ohkawa *et al.* 1998), in which a conspicuous increase in G_{ST} values was noted, reflecting a rapid decline in genetic diversity within populations (Nei 1973; Hamrick & Nason 1995). Such small, isolated lowland populations no longer exhibit typical L-shaped curves in size (or stage) structure, unlike normal mature populations in climax vegetation. However, we found no evidence of a decline in genetic diversity in the SERC beech populations.

If sufficient numbers of mature trees are present within a local population, a continuous supply of offspring is guaranteed, despite the fact that intermittent recruitments because of masting are usual in beeches, and high levels of genetic diversity will be maintained by frequent genetic recombination and gene reshuffling.

We would like to emphasize the fact that if we conduct customary random sampling and calculations on genetic polymorphism, disregarding different stage (or age) classes borne in different seasons extending over the past 50 years or more for perennial plant populations, and including woody plants such as beeches or long-lived perennial herbs (Kawano 1975; Kawano, 1985; Takada *et al.* 1998; Ohara *et al.* 2001), then the data we obtain would be a mixture of different generations borne in different seasons with a combination of different parental plants. In

the case of populations with overlapping generations, stage-class (or size-class) discriminations for genetic analyses provide critical information about the background of the reproductive activities in different seasons and the complex genetic structures of the population in question.

Based on our investigation, the SERC forest stands were heavily disturbed by lumbering in the past and many areas were cleared and used for farming (see Tibbs 1995). Despite this, over the past 150 years (since approximately 1850), the forest stands, presumed to be once dominated by American beech, have recovered and the forest communities are at present rich in species, including *Liriodendron tulipifera*, *Liquidambar styraciflua*, *Carya glabra* and *Cornus florida*. These woody species are all native elements of the typical coastal plain forest vegetation in the area (Brush *et al.* 1980; Rohrig & Ulrich 1991; Peters 1997). In addition, the present study showed that the species diversity associated with the understory of beech forest stands is rich and includes a number of elements indigenous to deciduous coastal plain forests, for example, *Podophyllum peltatum*, *Arisaema triphyllum* and *Uvularia perfoliata*. This fact clearly indicates that remnant forest stands in the surrounding areas have restored a good number of potential native elements to the woodland environments. Thus, the implications of the present study are not only on the recovery of American beech populations, a dominant species, but also on the associated elements in the natural

forest stands that once coexisted in this area. This is important evidence and indicates that even fragmented forest stands have the potential to restore species that coexisted in the forest vegetation.

Therefore, in terms of conservation biology, we could ascertain the time required for the recovery of major forest components during the successional process, and also of the other elements associated with the forest communities, for example, those which occur on the forest floor, such as hemicryptophytes and geophytes.

Although we have emphasized the levels of genetic diversity in the American beech forest stands currently existing in the area, the levels of genetic diversity in these beech populations were unexpectedly high, and also rich in vegetation and floristic composition.

Therefore, considering the present circumstances in forest communities as well as the floristic compositions, the degree of richness in the SERC beech forest stands clearly exhibits a rather swift recovery in American beech forests after abandonment. This is important evidence in terms of conservation biology because it highlights the significance of preserving fragmented populations of native elements remaining within forested areas.

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Appendix I

Allele frequencies for each size class

Locus	Allele	40 ≤ d.b.h. < 60 cm	60 ≤ d.b.h. < 80 cm	d.b.h. ≥ 80 cm	Total
<i>Lap</i>	<i>a</i>	0.059	0.082	0.130	0.068
	<i>b</i>	0.004	0.000	0.000	0.003
	<i>c</i>	0.348	0.409	0.283	0.354
	<i>d</i>	0.023	0.000	0.000	0.017
	<i>e</i>	0.565	0.509	0.587	0.557
<i>6Pgdh1</i>	<i>b</i>	0.998	1.000	1.000	0.998
	<i>c</i>	0.002	0.000	0.000	0.002
<i>6Pgdh2</i>	<i>a</i>	0.599	0.620	0.522	0.597
	<i>b</i>	0.399	0.380	0.478	0.401
	<i>c</i>	0.002	0.000	0.000	0.002
<i>6Pgdh3</i>	<i>a</i>	0.004	0.000	0.000	0.003
	<i>b</i>	0.527	0.509	0.478	0.521
	<i>c</i>	0.468	0.491	0.522	0.476
<i>Fum</i>	<i>a</i>	0.150	0.182	0.043	0.148
	<i>b</i>	0.778	0.755	0.913	0.784
	<i>c</i>	0.072	0.064	0.043	0.068
<i>Pgm</i>	<i>a</i>	0.002	0.000	0.000	0.002
	<i>b</i>	0.958	0.991	0.978	0.965
<i>Pgi</i>	<i>c</i>	0.040	0.009	0.022	0.033
	<i>b</i>	0.981	0.982	0.978	0.981
	<i>c</i>	0.015	0.018	0.000	0.014
	<i>d</i>	0.004	0.000	0.022	0.005
<i>Adh1</i>	<i>a</i>	0.025	0.027	0.000	0.024
	<i>b</i>	0.975	0.973	1.000	0.976
<i>Got1</i>	<i>a</i>	0.002	0.000	0.000	0.002
	<i>b</i>	0.998	1.000	1.000	0.998
<i>Got3</i>	<i>a</i>	0.002	0.000	0.000	0.002
	<i>b</i>	0.998	1.000	1.000	0.998
<i>Mdh</i>	<i>a</i>	0.004	0.000	0.000	0.004
	<i>b</i>	0.996	1.000	1.000	0.996
<i>Kth</i>	<i>a</i>	0.009	0.000	0.000	0.007
	<i>b</i>	0.991	1.000	1.000	0.993
<i>Aco</i>	<i>a</i>	0.994	0.987	0.950	0.991
	<i>b</i>	0.006	0.013	0.050	0.009
<i>Dia</i>	<i>b</i>	0.998	1.000	1.000	0.998
	<i>c</i>	0.002	0.000	0.000	0.002
<i>Acp</i>	<i>a</i>	0.753	0.736	0.783	0.752
	<i>b</i>	0.219	0.227	0.174	0.217
	<i>c</i>	0.025	0.036	0.043	0.029
	<i>d</i>	0.002	0.000	0.000	0.002

Appendix II

Genotypic frequencies for each size class

		40 ≤ d.b.h. < 60 cm	60 ≤ d.b.h. < 80 cm	d.b.h. ≥ 80 cm
<i>Lap</i>	<i>aa</i>	0.000	0.018	0.000
	<i>ac</i>	0.034	0.073	0.087
	<i>ad</i>	0.004	0.000	0.000
	<i>ae</i>	0.080	0.055	0.174
	<i>bc</i>	0.008	0.000	0.000
	<i>cc</i>	0.101	0.127	0.087
	<i>ce</i>	0.451	0.491	0.304
	<i>dd</i>	0.004	0.000	0.000
	<i>de</i>	0.034	0.000	0.000
	<i>ee</i>	0.283	0.236	0.348
<i>6Pgdh1</i>	<i>bb</i>	0.996	1.000	1.000
	<i>bc</i>	0.004	0.000	0.000
<i>6Pgdh2</i>	<i>aa</i>	0.325	0.389	0.174
	<i>ab</i>	0.544	0.463	0.696
	<i>ac</i>	0.004	0.000	0.000
	<i>bb</i>	0.127	0.148	0.130
<i>6Pgdh3</i>	<i>ab</i>	0.004	0.000	0.000
	<i>ac</i>	0.004	0.000	0.000
	<i>bb</i>	0.274	0.309	0.130
	<i>bc</i>	0.502	0.400	0.696
	<i>cc</i>	0.215	0.291	0.174
<i>Fum</i>	<i>aa</i>	0.030	0.055	0.000
	<i>ab</i>	0.219	0.236	0.087
	<i>ac</i>	0.021	0.018	0.000
	<i>bb</i>	0.608	0.582	0.826
	<i>bc</i>	0.122	0.109	0.087
	<i>cc</i>	0.000	0.000	0.000
<i>Pgm</i>	<i>ab</i>	0.004	0.000	0.000
	<i>bb</i>	0.920	0.982	0.957
	<i>bc</i>	0.072	0.018	0.043
	<i>cc</i>	0.004	0.000	0.000
<i>Pgi</i>	<i>bb</i>	0.962	0.964	0.957
	<i>bc</i>	0.030	0.036	0.000
	<i>bd</i>	0.008	0.000	0.043
<i>Adh1</i>	<i>ab</i>	0.051	0.055	0.000
	<i>bb</i>	0.949	0.945	1.000
<i>Got1</i>	<i>ab</i>	0.004	0.000	0.000
	<i>bb</i>	0.996	1.000	1.000
<i>Got3</i>	<i>ab</i>	0.004	0.000	0.000
	<i>bb</i>	0.996	1.000	1.000
<i>Mdh</i>	<i>ab</i>	0.009	0.000	0.000
	<i>bb</i>	0.991	1.000	1.000
<i>Idh</i>	<i>ab</i>	0.017	0.000	0.000
	<i>bb</i>	0.983	1.000	1.000
<i>Aco</i>	<i>aa</i>	0.991	0.974	0.900
	<i>ab</i>	0.004	0.026	0.100
	<i>bb</i>	0.004	0.000	0.000
<i>Dia</i>	<i>bb</i>	0.996	1.000	1.000
	<i>bc</i>	0.004	0.000	0.000
<i>Acp</i>	<i>aa</i>	0.591	0.564	0.652
	<i>ab</i>	0.287	0.309	0.174
	<i>ac</i>	0.034	0.036	0.087
	<i>ad</i>	0.004	0.000	0.000
	<i>bb</i>	0.076	0.073	0.087
	<i>cc</i>	0.008	0.018	0.000

Appendix III*Nei's estimation of heterozygosity and Wright's F-statistics*

Locus	<i>H_o</i>	<i>H_s</i>	<i>F_{it}</i>	<i>F_{st}</i>	<i>F_{is}</i>
<i>Lap</i>	0.598	0.566	−0.087	0.001	−0.089
<i>6pgdh1</i>	0.001	0.001	−0.004	−0.007	0.003
<i>6pdhg2</i>	0.569	0.489	−0.128	−0.002	−0.127
<i>6pgdh3</i>	0.535	0.506	−0.007	0.006	0.001
<i>Fum</i>	0.300	0.311	0.033	0.009	0.025
<i>Pgm</i>	0.046	0.048	0.063	0.003	0.060
<i>Pgi</i>	0.039	0.039	−0.017	−0.005	0.012
<i>Adh1</i>	0.035	0.035	−0.025	−0.003	0.022
<i>Got1</i>	0.001	0.001	−0.004	−0.007	0.003
<i>Got3</i>	0.001	0.001	−0.004	−0.007	0.003
<i>Mdh</i>	0.003	0.003	−0.009	−0.010	0.001
<i>Idh</i>	0.006	0.006	−0.010	−0.007	−0.003
<i>Aco</i>	0.044	0.046	0.400	0.009	0.394
<i>Dia</i>	0.001	0.001	−0.008	−0.011	0.003
<i>Acp</i>	0.310	0.386	0.159	−0.007	0.165
Overall	0.166	0.163	−0.013	−0.001	−0.012

Appendix IV

Significant values for the spatial autocorrelation, *I*(20)

Locus/allele	0-20	20-40	40-60	60-80	80-100	100-120	120-140	140-160	160-180	180-200	200-220	220-240	240-260	260-280	280-300	300-320	320-340	340-360	360-380	380-400	400-420	420-440	440-460	460-480	480-500
<i>Acp-a</i>																									
<i>Acp-b</i>																									
<i>Acp-c</i>	++			+	+							++									+				
<i>Idh</i>				+	+																	+			
<i>Aco</i>	+																								
<i>Adh1</i>	+																								
<i>Lap-a</i>	+	+	+	+	+	+																			
<i>Lap-b</i>	+	+	+	+	+	+																			
<i>Lap-c</i>	+	+	+	+	+	+																			
<i>Lap-d</i>	+	+	+	+	+	+																			
<i>Lap-e</i>	+	+	+	+	+	+																			
<i>6Pgdlh2-a</i>	+	+	+	+	+	+																			
<i>6Pgdlh2-b</i>	+	+	+	+	+	+																			
<i>6Pgdlh2-c</i>	+	+	+	+	+	+																			
<i>6Pgdlh3-a</i>	+	+	+	+	+	+																			
<i>6Pgdlh3-b</i>	+	+	+	+	+	+																			
<i>6Pgdlh3-c</i>	+	+	+	+	+	+																			
<i>Fum-a</i>	+	+	+	+	+	+																			
<i>Fum-b</i>	+	+	+	+	+	+																			
<i>Fum-c</i>	+	+	+	+	+	+																			
<i>Pgm-a</i>	+	+	+	+	+	+																			
<i>Pgm-b</i>	+	+	+	+	+	+																			
<i>Pgm-c</i>	+	+	+	+	+	+																			
<i>Pgi-b</i>	+	+	+	+	+	+																			
<i>Pgi-c</i>	+	+	+	+	+	+																			
<i>Pgi-d</i>	+	+	+	+	+	+																			

Blank, not significant; +, positive significant value at $P > 0.05$; ++, positive significant value at $P > 0.01$; --, negative significant value at $P > 0.05$; ---, negative significant value at $P > 0.01$.